County Median Income and Diabetes-Related Hospitalizations*

Median Adjusted Gross Income and Diabetes Composite Hospitalization Rates in California, 2022

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Diabetes is a major public health concern, and socioeconomic factors may be associated with differences in diabetes composite observed hospitalization rates. This study examines the relationship between median adjusted gross income and diabetes composite observed hospitalization rates across California counties. We combined 2022 county-level diabetes hospitalization data with 2022 median income data from the California Franchise Tax Board. A simple linear regression model was used to evaluate the association between income and diabetes composite observed hospitalization rates across 58 counties. The results showed a statistically significant negative association between median adjusted gross income and the Diabetes Composite observed hospitalization rate. These findings suggest that higher-income counties tend to have lower diabetes composite observed hospitalization rates. However, because this was an observational county-level analysis, the results demonstrate an association rather than a causal relationship.

1 Introduction

Diabetes is a major public health issue in the United States. According to the Centers for Disease Control and Prevention (CDC), an estimated 40.1 million people in the United States had diabetes in 2023, representing approximately 12.0% of the U.S. population.[1] Diet, physical activity, and lifestyle are some of the factors commonly associated with diabetes. In this project, we focus on income as a socioeconomic factor and examine its relationship with diabetes-related preventable hospitalization rates across California counties in 2022. Specifically, we use linear regression analysis to investigate whether county-level median adjusted gross income is associated with the Diabetes Composite observed hospitalization rate.

*Project repository available at: https://github.com/seynLi77/MATH261A-Project1-Diabetes.

Our research question is:

Is median adjusted gross income associated with the Diabetes Composite observed hospitalization rate across California counties in 2022?

We hypothesize that counties with higher median adjusted gross income will have lower Diabetes Composite observed hospitalization rates. Therefore, we expect to observe a negative relationship between income and diabetes-related preventable hospitalizations across California counties.

Section 2 introduces the two data sources, the observational unit, the data-cleaning and merging steps, and descriptive statistics. Section 3 describes the regression model, hypotheses, and assumptions. Section 4 presents the fitted model and diagnostic checks, and Section 5 interprets the findings and limitations.

2 Data

The observational unit is one California county in 2022. The analysis combines two county-level datasets from California Open Data. First, the California Franchise Tax Board's `2023-b-6-comparison-by-county.csv` provides median adjusted gross income across all filing statuses for each county and taxable year. Second, the California Department of Health Care Access and Information's (HCAI) `ca-hcai-preventablehospitalizations-county_2005-2024.csv` provides observed rates for preventable hospitalization measures by county and year.

The `2023-b-6-comparison-by-county.csv` provides personal income tax statistics for taxpayers by the county of residence based on their tax returns. These data were collected annually by the California Franchise Tax Board. Each row describes county-level tax statistics for a particular taxable year. The analysis focuses on the `Median Income` variable for 2022 in the dataset, which represents median adjusted gross income for taxable and non-taxable returns for all filing statuses.

The `ca-hcai-preventablehospitalizations-county_2005-2024.csv` provides hospitalization counts and rates, statewide and by county, for 10 ambulatory care sensitive conditions plus 4 composite measures. These data were collected by the California Department of Health Care Access and Information (HCAI). Hospitalizations due to these medical conditions are potentially preventable through access to high-quality outpatient care. Each row of data describes the hospitalization information for a California county in a given year.

Our analysis focuses on records classified as `Diabetes Composite` in the `PQIDescription` column and uses `ObsRate_ICD10` for 2022. This variable is the response variable in our analysis. It measures the number of diabetes-related preventable hospitalizations per 100,000 county residents, rather than diabetes prevalence or the total number of hospitalizations.

Although the hospitalization outcome is reported as a rate per 100,000 population, counties vary in population size. The rates for smaller counties may be more variable because they are

affected more by a small number of hospitalization events. Since each county has equal weight in the regression, smaller counties may have a greater effect on the estimated relationship than their population size would suggest.

The diabetes and income datasets were first filtered to retain observations from 2022. From the diabetes dataset, only records corresponding to the Diabetes Composite measure were retained, along with the county name and the ICD-10 observation rate. From the income dataset, the county name and median income were selected.

The two datasets were then merged using an inner join based on the County variable. This ensured that the regression dataset included only counties with available information in both datasets. Finally, median income was divided by 1,000 to express the variable in thousands of dollars, making the regression coefficients easier to interpret.

No observations were manually deleted, imputed, or otherwise modified beyond these data preparation steps. Therefore, no additional data cleaning procedures were necessary for this analysis.

Table 1: Summary statistics for the variables used in the analysis.

Variable	N	Mini-mum	Me-dian	Mean	Maxi-mum
Median adjusted gross income	58	32.52	46.73	49.88	84.29
Diabetes Composite observed hospitalization rate	58	0.00	189.35	192.35	340.90

Table 1 summarizes the descriptive statistics for the variables used in the regression analysis. The final analytic dataset contains 58 California counties. Median adjusted gross income ranges from approximately $32.52 thousand to $84.29 thousand, with a median of approximately $46.73 thousand. The Diabetes Composite observed hospitalization rate ranges from 0 to 340.9 hospitalizations per 100,000 population, with a median of 189.35.

Figure 1 suggests a negative linear relationship between median adjusted gross income and the Diabetes Composite observed hospitalization rate.

3 Methods

We fit a simple linear regression model to examine the relationship between median adjusted gross income and diabetes-related preventable hospitalization rates across California counties:

$$Y_i = \beta_0 + \beta_1 X_i + \varepsilon_i$$

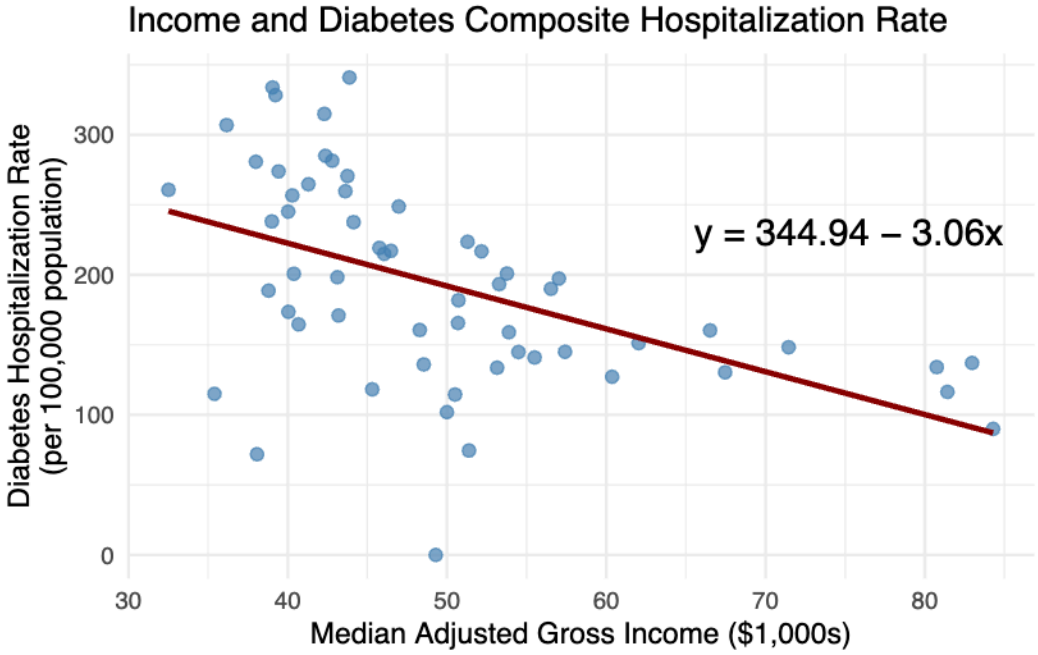


Figure 1: Scatter plot of median adjusted gross income, measured in $1,000s (x-axis), and the Diabetes Composite observed hospitalization rate per 100,000 population (y-axis), with the fitted linear regression line.

Here, Y_i is the Diabetes Composite observed hospitalization rate for county i, X_i is the county's median adjusted gross income in \$1,000s, β_0 is the intercept, β_1 is the slope, and ε_i is the error term. The intercept represents the predicted hospitalization rate when median income is zero. The slope represents the expected change in the hospitalization rate associated with a \$1,000 increase in median adjusted gross income.

The error terms are assumed to have mean zero, constant variance σ^2, and to be independent across counties. For the usual small-sample t-test and confidence interval, the errors are also assumed to be approximately normally distributed. We assess linearity and constant variance with the scatter plot and residual plot, and assess normality with a Q-Q plot. These assumptions are approximations because counties may differ in population size, healthcare access, demographics, and other factors not included in the model.

The hypotheses for the slope are $H_0 : \beta_1 = 0$ versus $H_A : \beta_1 \neq 0$. We use the statistic $t = (\hat{\beta}_1 - 0)/\,\mathrm{SE}(\hat{\beta}_1)$ with $n - 2$ degrees of freedom and a two-sided significance level of $\alpha = 0.05$.

The analysis uses county-level data from 2022. Counties are matched by name, and only counties with available observations in both datasets are included.

All statistical analyses were performed using R.[4]

4 Results

Table 2: Regression coefficient estimates and test statistics.

Term	Estimate	Std..Error	t.value	p.value	CI.Lower	CI.Upper
(Intercept)	344.941	35.498	9.717	0	273.830	416.052
MedianIncome_k	-3.059	0.692	-4.422	0	-4.445	-1.673

Table 2 summarizes the estimated coefficients, standard errors, test statistics, p-values, and 95% confidence intervals for the fitted regression model.

The estimated slope is $\hat{\beta}_1 =$ -3.06. This means that a \$1,000 increase in median adjusted gross income is associated with an estimated decrease of approximately 3.06 hospitalizations per 100,000 population in the Diabetes Composite observed hospitalization rate.

The estimated intercept is $\hat{\beta}_0 = 344.94$. This represents the predicted Diabetes Composite observed rate when median adjusted gross income equals zero. However, because an income of zero is outside the meaningful range of the data, the intercept is mainly a mathematical component of the regression equation.

We use a two-sided t-test to compare the null hypothesis $H_0 : \beta_1 = 0$ with the alternative hypothesis $H_A : \beta_1 \neq 0$.

The test statistic is t = -4.42, with a p-value of 4.56e-05.

The 95% confidence interval for the slope is (-4.44, -1.67). This provides evidence of a statistically significant negative association between median adjusted gross income and the Diabetes Composite observed hospitalization rate.

The scatter plot in Figure 1 suggests an approximately negative linear relationship between the two variables. However, the scatter plot alone cannot fully assess all regression assumptions. Therefore, residual plots should also be examined to evaluate the assumptions of constant variance and normality.

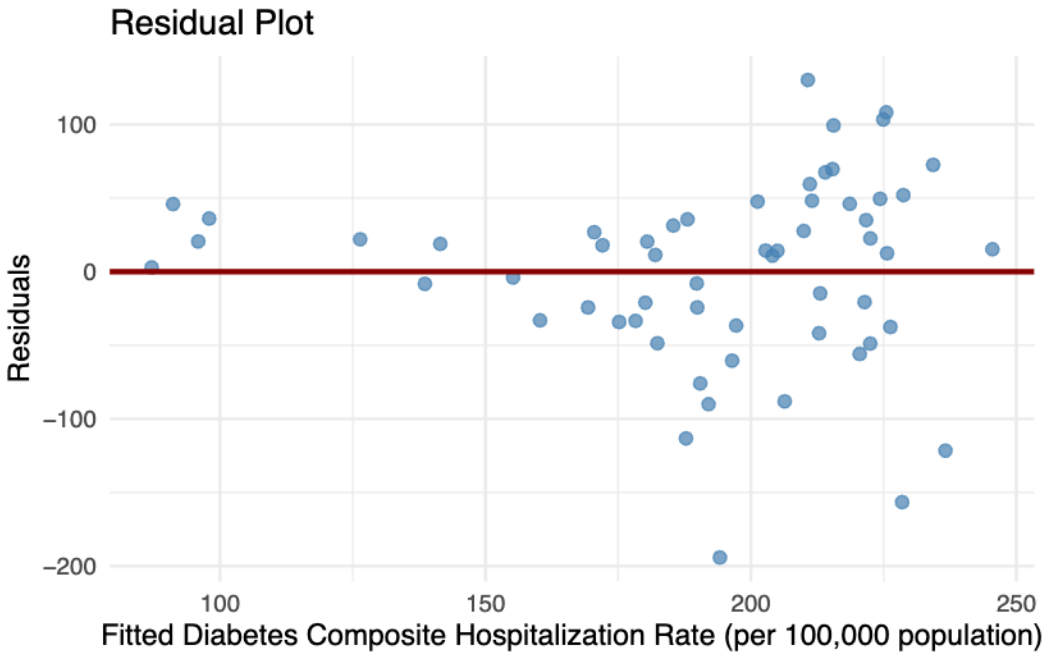


Figure 2: Residuals plotted against fitted values for the simple linear regression of the Diabetes Composite observed hospitalization rate on median adjusted gross income measured in $1,000s.

Figure 2 shows that the residuals are generally centered around zero, but they are not evenly spread across all fitted values. The residuals appear to have a larger spread for higher fitted values, suggesting that the constant-variance assumption may not be fully satisfied. If the variance is not constant, the estimated standard errors may be inaccurate, which could affect the reported t-test and confidence interval. There are also several relatively large residuals that may represent unusual counties and could affect the fitted regression line. However, their influence would need to be assessed using additional measures, such as Cook's distance.

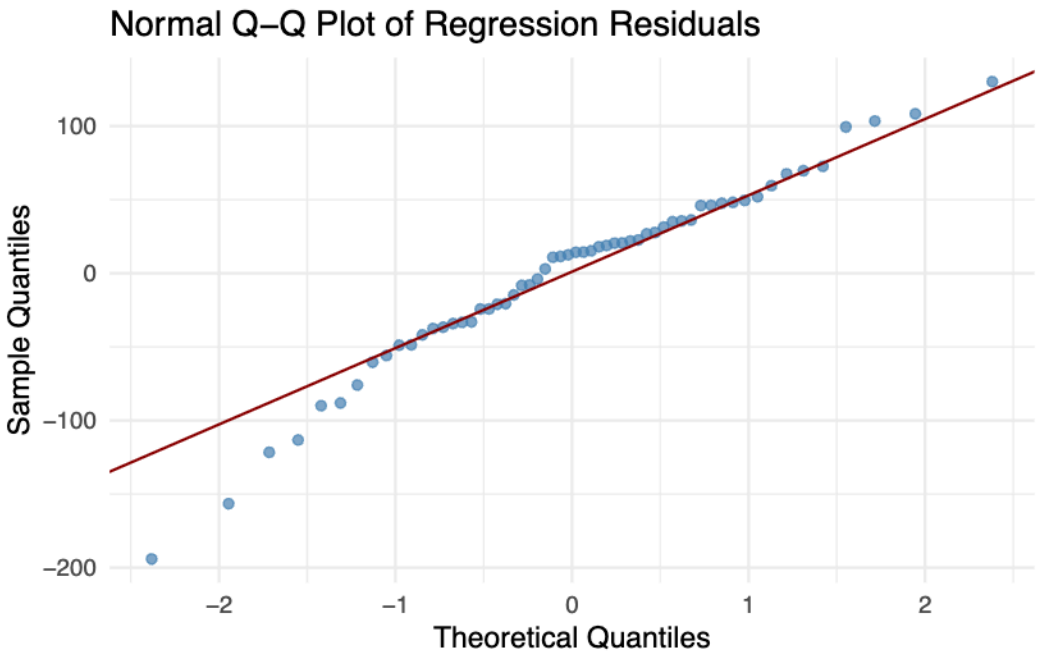


Figure 3: Normal Q-Q plot of the residuals from the simple linear regression of the Diabetes Composite observed hospitalization rate on median adjusted gross income.

Figure 3 shows that the residuals are relatively close to the reference line in the middle of the distribution, but they deviate from the line in both tails, especially in the lower tail. This suggests that the residuals are not perfectly normally distributed and may contain outlying observations or heavier tails than a normal distribution. Together with the possible violation of the constant-variance assumption described above, this suggests that the usual inferential results should be interpreted with caution.

Overall, the diagnostic plots suggest that the linear model provides a useful summary of the association, but some assumptions may be violated. The possible nonconstant variance means that the usual standard errors, p-value, and confidence interval may be inaccurate. Deviations from normality may also affect the small-sample t-test, especially in the presence of extreme or influential counties. Therefore, the estimated negative slope should be interpreted as evidence of an association, while the reported inferential results should be treated with caution. A follow-up analysis could use heteroscedasticity-robust standard errors and formally assess influential observations.

5 Discussion

This analysis examined whether median adjusted gross income was associated with the Diabetes Composite observed hospitalization rate across California counties in 2022. The regression results showed a statistically significant negative association: counties with higher median adjusted gross income tended to have lower Diabetes Composite observed hospitalization rates. Therefore, the results provide evidence consistent with our hypothesis. However, because this was an observational analysis, the results describe an association and do not establish that income causes differences in hospitalization rates.

Several factors could plausibly contribute to this association. Higher-income counties may differ in access to healthy food, recreational opportunities, preventive care, and other social determinants of health. Higher-income areas may also have better access to healthcare resources, allowing residents to detect and manage health problems earlier. However, these factors were not directly measured in this analysis, so they should be interpreted as possible explanations rather than demonstrated causal mechanisms.

One limitation is the presence of unusual observations. For example, Alpine County has a Diabetes Composite observed rate of 0.0 while its median income is $49,293. Because Alpine County has a very small population, its observed rate may be unstable and sensitive to a small number of hospitalization events. Modoc County also has relatively low income and a relatively low observed hospitalization rate, which does not follow the overall negative relationship. These counties may have a disproportionate influence on the fitted regression line.

Another limitation is that this is a county-level analysis. Because the analysis is based on aggregate county-level data, the observed association cannot be interpreted as evidence that individuals with higher incomes are less likely to have diabetes or less likely to experience a diabetes-related preventable hospitalization. This is an example of the ecological limitation of using group-level data to make individual-level conclusions.

In addition, the outcome is a hospitalization rate rather than a measure of diabetes prevalence. Hospitalization rates may reflect both underlying health conditions and differences in access to, use of, and admission to healthcare services. Other factors, including population age distribution, race and ethnicity, insurance coverage, healthcare access, population size, and other unmeasured characteristics, may also confound the association.

Future analyses could use population-weighted methods, include additional county-level covariates, and investigate influential counties more formally. Using data from multiple years could also help determine whether the negative association is consistent over time. Individual-level data would be needed to examine whether income is associated with an individual's diabetes status or risk of diabetes-related hospitalization.

6 References

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[4] R Core Team. (2026). R: A Language and Environment for Statistical Computing. R Foundation for Statistical Computing, Vienna, Austria. https://www.R-project. org/